Letter Sequences & Alphabet Code (Set B)

Year 4 Pt 2 · **Mark Scheme**

Lesson Worksheet — Answers

Spotting the Pattern

1. I — the gap between each letter is +2 (A, C, E, G, I).
2. V — the gap is -1 going backwards (Z, Y, X, W, V).
3. B (6) — F is the 6th letter of the alphabet.
4. A, B, C, D — C–E, G–I, M–O and P–R are all 2 apart; T and U are only 1 apart.
5. N — the gap is +3 throughout (B, E, H, K, N).
6. C (O) — O is the 15th letter of the alphabet.
7. Q — the gap is -2 throughout (Y, W, U, S, Q).
8. C — A, C, E, G, I (skipping one letter each time).
9. B (E) — shifting D forward by 1 gives E.
10. A, B, D, E — the gap is consistently 3, so the next letter after O is R, and the first letter given is C; the sequence moves forwards (not backwards), so C is false.
11. L — the gap is +2 throughout (D, F, H, J, L).
12. C (20) — T is the 20th letter of the alphabet.
13. I — the gap is -2 throughout (Q, O, M, K, I).
14. B (F) — shifting H back by 2 gives F.
15. A, C, E — P, R and W all come after M in the alphabet; J and L come before M.

Applying the Pattern

16. N — gaps alternate +2, +3, +2, +3: B→D(+2), D→G(+3), G→I(+2), I→L(+3), L→N(+2).
17. GEF — C+4=G, A+4=E, B+4=F.
18. S — gaps increase by 1 each time: A→D(+3), D→H(+4), H→M(+5), M→S(+6).
19. B (O) — 7+8=15, and the 15th letter is O.
20. A, B, D, E — all have a consistent gap of 4; C has a gap of 3.
21. N — the gap is consistently -3: Z, W, T, Q, N.
22. DFC — shifting back 3: G→D, I→F, F→C.
23. D — this interleaves A,B,C,D… (forwards) with Z,Y,X… (backwards): A, Z, B, Y, C, X, so the next term is D (continuing the forward sequence), with X being the paired backward term; accept D as the next single term shown.
24. B (T) — 26−6=20, and the 20th letter is T.
25. A, B, C, D, E — every pair is correctly shifted forward by 5, including Z→E which correctly wraps round past the end of the alphabet.
26. U — gaps increase by 1 each time: C→F(+3), F→J(+4), J→O(+5), O→U(+6).
27. Z — W is the 23rd letter; 23+3=26, which is Z.
28. Z — gap is -3 throughout: L, I, F, C, Z (wrapping: C is 3rd letter, 3−3=0, which wraps to 26=Z).
29. A, C, D — the gap is +3 throughout, E is the first letter, and the gap between K and N is 3, not 2, so E is false; the sequence is increasing, not decreasing, so B is false.
30. Z — S is the 19th letter; 19+7=26, which is Z.

Codes in Context

31. Q — the gaps increase by 1 each time: B→C(+1), C→E(+2), E→H(+3), H→L(+4), L→Q(+5).

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32. The shift is +5 (A→F, B→G, etc). Y (25th letter) +5 = 30, which wraps round to 30−26=4, giving D. Z (26th) +5 = 31, wraps to 31−26=5, giving E. So "YZ" decodes to "DE".

33. 6 — C(3) + A(1) + B(2) = 6.

34. A, D, E — the gaps are -2, -3, -4 (consistent with a widening pattern) and the sequence decreases through the alphabet (not increases, so C is false). Following the pattern (-5 next), Q(17)−5=12, which is L, not N, so B is false.

35. H — D is the 4th letter; 4×2=8, and the 8th letter is H.

36. G and W — this interleaves A,C,E,G… (forward, skipping one) with Z,X,V,T… (backward, skipping one): A,Z,C,X,E,V,G,T — but matching the given pattern A,Z,C,X,E,_,_ the next two terms are G (forward sequence) and V (backward sequence).

37. D — the shift from F to K is +5; applying +5 to Y (25th letter): 25+5=30, wraps to 30−26=4, which is D.

38. The shift is +3 (P→S is +3, E→H is +3, N→Q is +3). Applying -3… decoding VQI: V(22)−3=19=S, Q(17)−3=14=N, I(9)−3=6=F. So "VQI" decodes to "SNF".

39. A, B, C, E — all describe the sequence correctly; D is false because the next letter after P (16th) adding 3 gives 19, which is S, not R.

40. XZY — reversing the alphabet: C(3rd) ↔ X(24th, since 27−3=24), A(1st) ↔ Z(26th), B(2nd) ↔ Y(25th). So "CAB" becomes "XZY".

41. P — the gaps increase by 1 each time: A→B(+1), B→D(+2), D→G(+3), G→K(+4); the next gap is +5, so K(11)+5=16, which is P.

42. DEI — D is the 4th letter (position 3 using the 0-indexed code), E is the 5th letter (code 4), I is the 9th letter (code 8).

43. CAT has vowel A only; shifting A forward by 2 gives C. Consonants C and T stay the same. Result: "CCT".

44. A, B, C — these confirm the consistent -3 gap and correctly calculate that the next term should be K (11th letter), not H; D and E are therefore incorrect, since H is NOT the correct next letter.

45. TDC — Step 1 (shift +2): B→D, A→C, T→V, giving "DCV". Step 2 (reverse the order of letters): "DCV" reversed is "VCD". So the final result is "VCD".

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Homework Worksheet — Answers

Quick Practice

1. J — gap +2 throughout (B, D, F, H, J).

2. O — gap -2 throughout (W, U, S, Q, O).

3. C (11) — J is the 11th letter.

4. A, B, D — B–E, F–I and N–Q are each 3 apart; K–L and R–S are only 1 apart.

5. M — gap +3 throughout (A, D, G, J, M).

6. C (T) — T is the 20th letter.

7. M — gap -3 throughout (Y, V, S, P, M).

8. D — B, D, F, H, J (skipping one letter each time).

9. D (G) — shifting E forward by 2 gives G.

10. A, C, D — the gap is consistently 3 throughout (A); the next letter after P (16+3=19) is S (C); the first letter is D (D). The sequence increases, not decreases, so B is false; the gap between J and M is 3, not 4, so E is false.

11. M — gap +2 throughout (E, G, I, K, M).

12. A (13) — N is the 13th letter.

13. L — gap -2 throughout (T, R, P, N, L).

14. B (F) — shifting G back by 1 gives F.

15. A, C, E — G, C and F all come before J; M and L come after J.

Mixed Practice

16. O — gaps increase by 1: A→C(+2), C→F(+3), F→J(+4), J→O(+5).

17. GRJ — D+3=G, O+3=R, G+3=J.

18. T — gaps increase by 1: B→E(+3), E→I(+4), I→N(+5), N→T(+6).

19. C (T) — 9+10=19, and the 19th letter is T.

20. A, B, D — gap of 5 throughout; C has a gap of 5 then jumps to X (gap 6) — checking C,H,M,R,X: C(3)H(8)M(13)R(18)X(24), gaps are 5,5,5,6, so C is false; E has gap 4, so false. Correct: A, B, D.

21. M, J — gaps of -3: Y, V, S, P, M, J.

22. FMI — J−4=F, Q−4=M, M−4=I.

23. G — this interleaves A,C,E,G… (forward, skip one) with Y,W,U… (backward, skip one): A,Y,C,W,E,U,G — matching the pattern shown (A,Y,C,W,E,__), the next term in the forward sequence is G.

24. B (P) — 26−10=16, and the 16th letter is P.

25. A, B, C, D, E — all are correctly shifted forward by 6, including U→A and Z→F which both wrap correctly.

26. V — gaps increase by 1: D→G(+3), G→K(+4), K→P(+5), P→V(+6).

27. B — X is the 24th letter; 24+4=28, wraps to 28−26=2, which is B.

28. B — gap -3 throughout: N, K, H, E, B.

29. A, C, D — gap is +3 throughout and F is the first term; the next letter after R (+3) is U, so C is true; the sequence increases, not decreases, so B is false; gap between L and O is 3, not 2, so E is false. Correct: A, C, D.

30. T — N is the 14th letter; 14+6=20, which is T.

Standard / Advanced

31. R — gaps increase by 1 each time: C→D(+1), D→F(+2), F→I(+3), I→M(+4), M→R(+5).

32. AB — shift +6: U(21)+6=27, wraps to 27−26=1=A. V(22)+6=28, wraps to 28−26=2=B. So "UV" decodes to "AB".

33. 26 — D(4) + O(15) + G(7) = 26.

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34. A, D, E — gaps are -3, -4, -5, consistent widening pattern (A true); gap between V and R is 4 (true, D); gap between R and M is 6 (true, E). The sequence decreases, not increases, so C is false. Following the pattern (next gap -6), M(13)−6=7, which is G, not I, so B is false.

35. I — C is the 3rd letter; 3×3=9, and the 9th letter is I.

36. H and U — interleaving B,D,F,H… (forward, skip one) with Y,W,U… (backward, skip one): B,Y,D,W,F,U,H — matching the given pattern (B,Y,D,W,F,__,__), the next two terms are U (backward sequence continuing from W) and H (forward sequence); giving the terms in order: U then H.

37. D — the shift from H to O is +7; applying +7 to W (23rd letter): 23+7=30, wraps to 30−26=4, which is D.

38. The shift is +3 (C→F, A→D, T→W, all +3). Decoding LWS with -3: L(12)−3=9=I, W(23)−3=20=T, S(19)−3=16=P. So "LWS" decodes to "ITP".

39. A, B, C, E — gap +3 throughout (A); every 3rd letter starting at E (B); position numbers 5,8,11,14,17 (C); the gap is consistent throughout (E). The next letter after Q (17+3=20) is T, not S, so D is false.

40. WLT — reversing the alphabet: D(4th) ↔ W(23rd, since 27−4=23), O(15th) ↔ L(12th, since 27−15=12), G(7th) ↔ T(20th, since 27−7=20). So "DOG" becomes "WLT".

41. Q — gaps increase by 1 each time: B→C(+1), C→E(+2), E→H(+3), H→L(+4), L→Q(+5).

42. BNE — B is the 2nd letter (code 1), N is the 14th letter (code 13), E is the 5th letter (code 4).

43. DOG has vowels O only; D and G are consonants and shift forward by 1: D→E, G→H. O stays unchanged. Result: "EOH".

44. A, B, C — these confirm a consistent -3 gap and correctly calculate the true next term as J (10th letter); D and E are incorrect since G is NOT the correct next letter — the full sequence must be used, not just the last term.

45. OND — Step 1 (shift back 1): D→C, O→N, G→F, giving "CNF". Step 2 (reverse the order): "CNF" reversed is "FNC". So the final result is "FNC".

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